

EDSD 2024–2025: Sequence Analysis’ Assignment

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1 Introduction

For this assignment, we will work with data provided to illustrate life course trajectories of a sample of individuals from the German Family Panel (pairfam) data.

We will work with four different partnership states to construct the trajectories:

- *S*: Single
- *LAT*: Living Apart Together
- *COH*: Cohabiting
- *MAR*: Married

Once we identified some missing values either for the states or for individual weights, we opted to exclude these individuals of our sample.

We ended up with 1727 individuals to compute the sequences.

2 Import the data in R and describe the distribution of life states using relevant sequence analysis tools, including tables, indexes and figures.

First step is to create the dataset based on the sequences to describe the distribution of life states.

To describe the distribution of life course states, we can use different measures.

1. Time spent in different states

	Mean	Var	Stdev	SE
S	4.2	1.50	1.22	0.029
LAT	2.2	1.73	1.32	0.031
COH	1.6	0.86	0.93	0.022
MAR	4.3	0.47	0.68	0.016

2. Number of transitions

```
[1] 11.37856
```

3. Transition matrix between states

	[-> S]	[-> LAT]	[-> COH]	[-> MAR]
[S ->]	0.941915367	0.025436702	0.002844205	0.029803726
[LAT ->]	0.021144585	0.949870848	0.021469088	0.007515479
[COH ->]	0.007410091	0.002942081	0.972258815	0.017389013
[MAR ->]	0.025729730	0.002619736	0.003630421	0.968020113

4. State distribution at different ages (measured in months since the individuals started to be followed)

	id	state7	state55	state103	state151	state199	state247
S	0	0.57696693	0.3437856	0.2126568	0.14652223	0.1191562	0.12086659
LAT	0	0.34435576	0.2907640	0.1864310	0.09806157	0.0718358	0.05131129
COH	0	0.06100342	0.2291904	0.2759407	0.24458381	0.1584949	0.12029647
MAR	1	0.01767389	0.1362600	0.3249715	0.51083238	0.6505131	0.70752566

3 Calculate life course dissimilarities in the data

To compute the life course dissimilarities based on the data we are using, we applied a data-driven approach to compute the indel and similarity matrix, based on the transition matrix to assign the transition costs.

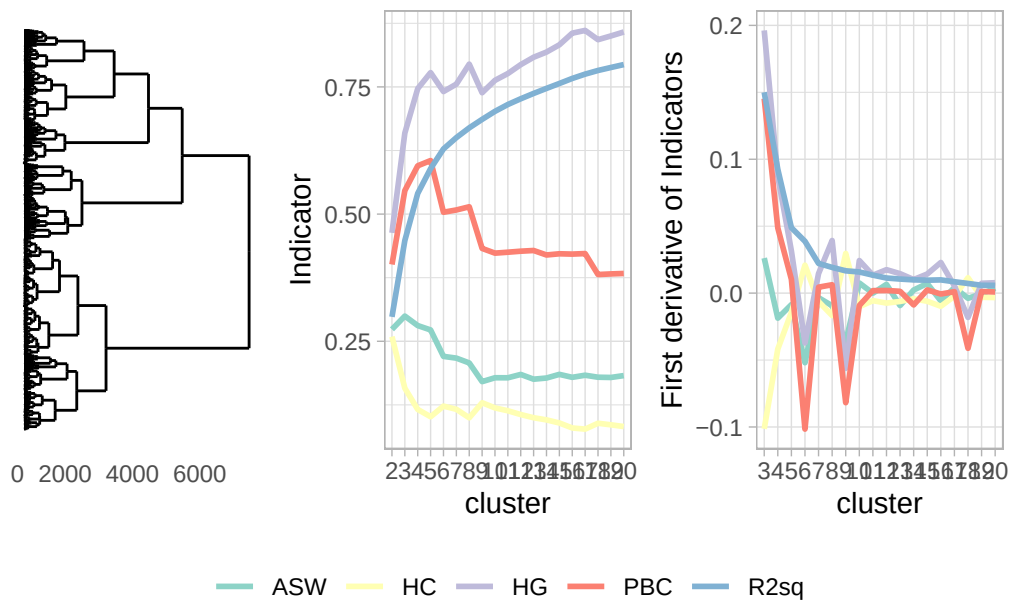
```
# we will use a not arbitrary indel value based on the transition matrix
trcost <- seqcost(df.seq, method="TRATE")

# calculating distance matrix

dist.om <- seqdist(
  df.seq,
  method="OM",
  indel=trcost$indel,
  sm=trcost$sm
)
```

4 Use a clustering methods to derive typical trajectories and describe groups

First, let's analyze the clustering distance distribution based on different number of clusters.



Sequence Analysis' Assignment

Based on the statistic indicators PBC and HG, the best number of clusters to be chosen is 5.

To describe the groups, let's plot the distribution of trajectories, based on each cluster.

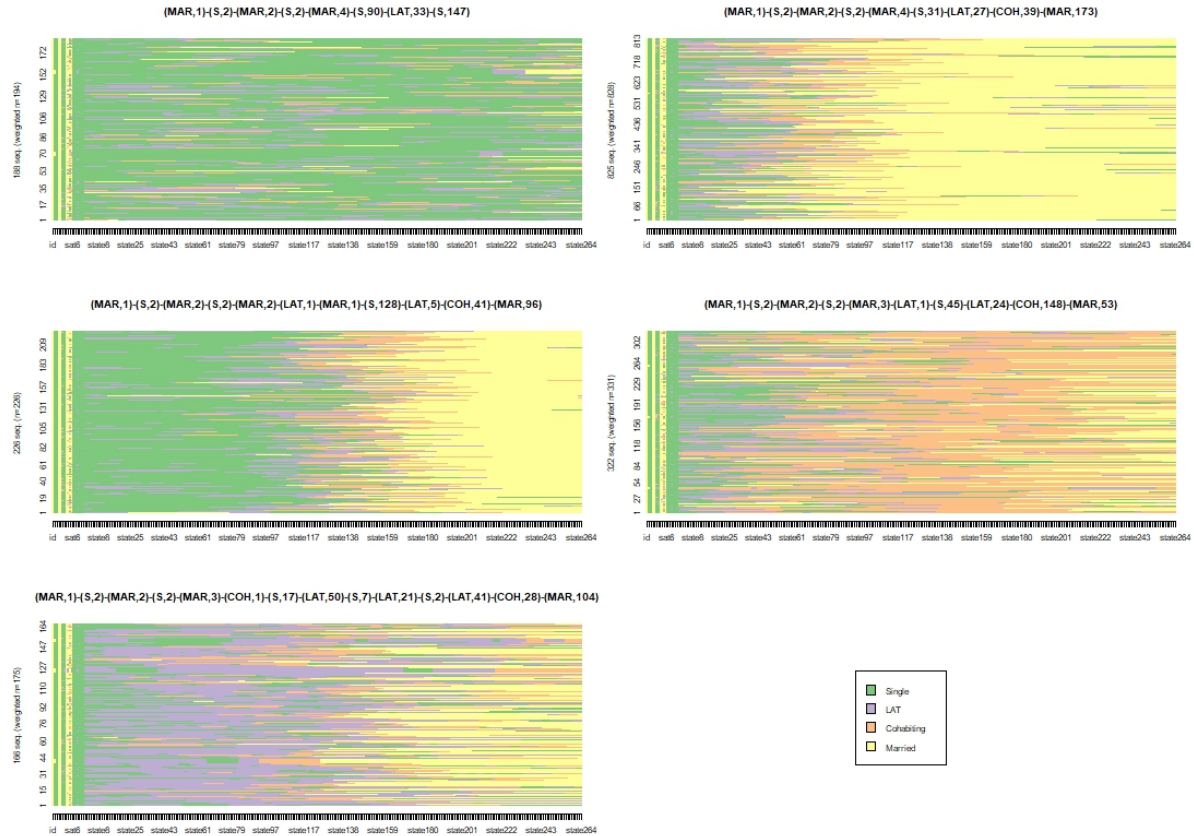


Figure 1: Trajectories for each cluster

5 (Optional) Describe group differences in the the distribution of life course clusters

For describing group differences in the distribution of life course by cluster, we can use different measures. Here we will use the distribution of states over life course.

- Distribution of states over life course by cluster

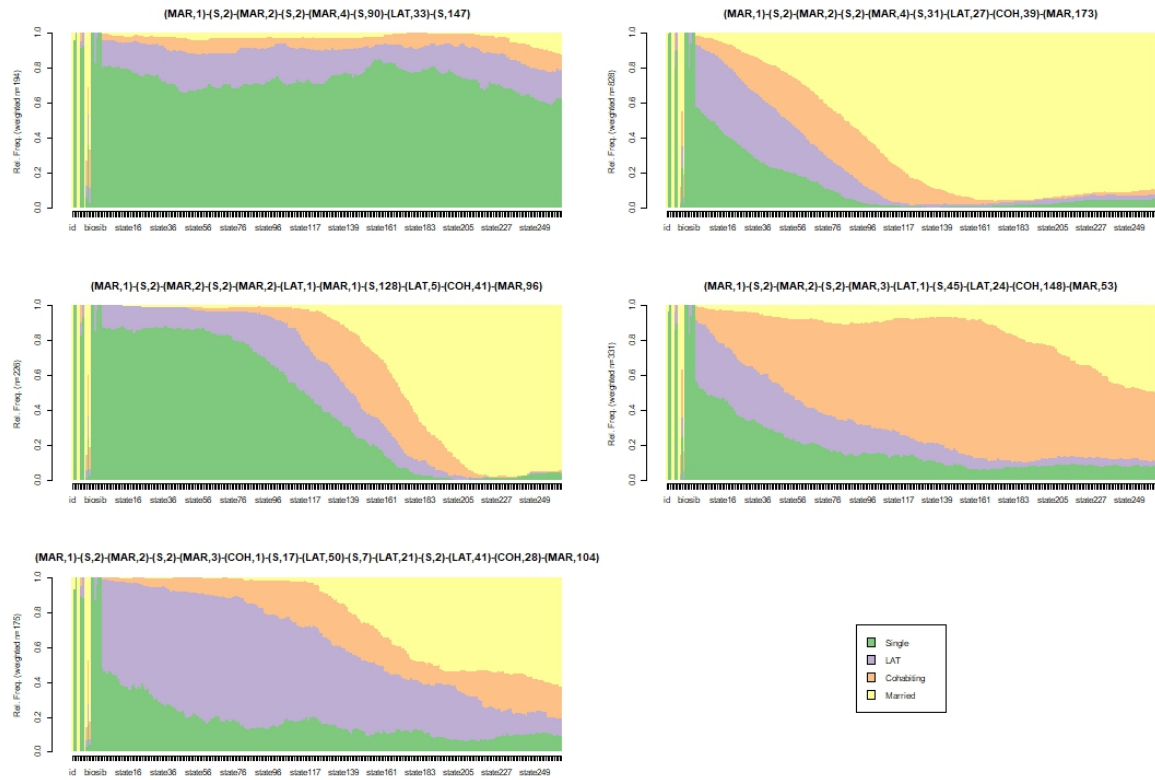


Figure 2: Distribution of states over life course by cluster